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Capacitated Agriculture Fleet Vehicle Routing with Implements and Limited Autonomy: A model and a two-phase solution approach

Aitor López-Sánchez[1], Marin Lujak[2*], Frederic Semet[3] and Holger Billhardt[4]

***Abstract*— In this paper, we study the vehicle routing problem (VRP) for a fleet of cooperative autonomous agricultural robots (agribots) equipped with detachable implements, with the goal of efficiently and sustainably completing agricultural tasks in precision crop farming. State of the art in the area of agribot fleet routing with detachable implements is lacking. Consequently, we propose the Capacitated Agriculture Fleet Vehicle Routing Problem with Implements and Limited Autonomy (CAFVRPILA), designed to optimize the agribot fleet's routes across a set of given agricultural tasks while considering implement capacities, agribot-implement compatibilities, and agribots' limited battery autonomies. A heuristic two-phase decomposition approach is proposed for this problem. Simulation experiments show that minimizing travel distances and costs with CAFVRPILA enhances sustainable farming while maximizing productivity and resource use. The results also demonstrate that synchronizing multiple operations improves efficiency, particularly in larger fleets.**

***Index Terms*— Path Planning for Multiple Mobile Robots or Agents; Agent-Based Systems; Robotics and Automation in Agriculture and Forestry; Vehicle Routing.**

I. Introduction

An autonomous agriculture mobile robot (agribot) or an autonomous tractor is already a commercial product, yet operational management solutions for the fleets including such tractors are lacking or inexistent. From the theoretic perspective, state of the art agricultural Vehicle Routing Problems primarily address homogeneous tractor fleets with a single type of agriculture operation and one type of crop. Modern cooperative agriculture fleets, however, already involve diverse types of autonomous tractors (agribots), each one requiring an attached implement (tool) to perform a specific agriculture operation, such as plowing, seeding, spraying, fertilization, fumigation, and harvesting. In an agriculture cooperative, these operations involve various agricultural fields belonging to the farmers composing the same. In the realm of this work, we consider an agriculture task to be an agriculture operation type on a specific field and with a specific resource demand. Efficient real-time coordination of routes of heterogeneous tractors and/or agribots (vehicles) owned by different farmers in a cooperative fleet is essential to adapt to unforeseen events for optimized efficient and fair resource allocation, task execution, and routing.

To this purpose, we introduce the Capacitated Agriculture Fleet Vehicle Routing Problem with Implements and Limited Autonomy (CAFVRPILA), an extension of the Agriculture Fleet Vehicle Routing Problem (AFVRP), designed to address the unique coordinating challenges of agriculture operations. It focuses on determining the most effective and efficient routes for an agribot fleet, taking into account agribot autonomy and compatibility with the implements and operations required to complete pending tasks.

The paper is organized as follows: Section II reviews the state of the art. Section III introduces the CAFVRPILA model, incorporating key factors such as implement capacities, vehicle-implement and implement-operation compatibilities, and limited agribot autonomies. In Section IV, we propose a solution approach based on a heuristic two-phase decomposition per operation. Section V presents simulation experiments, results, and related discussion. Our experiments demonstrate that the CAFVRPILA model, combined with the proposed heuristic solution, effectively minimizes travel distances and costs while seamlessly coordinating implement exchanges and operations. As a result, the generated routes optimize resource use and fleet productivity while reducing battery consumption and carbon emissions, promoting sustainable farming practices. Section VI concludes with a discussion of the results and directions for future work.

II. Related work

Traditionally, the AFVRP research has predominantly concentrated on scenarios involving a homogeneous fleet of vehicles performing single-operation crop-related tasks within a farm, aiming for the best cost savings [1]. An overview of the AFVRP family of problems can be found in [2]. State of the art generally assumes that each vehicle is equipped with a compatible implement. Thus, a combined tractor-implement unit can perform any task independently, without the need for switching between compatible implements [3], [4]. In addition, these works have considered various optimization criteria, such as input costs, minimization of non-working distance or time, and individual vehicle characteristics, including speed and fuel consumption [5], [6], [7]. Lastly, Utamima et al. extend the Agricultural Route Planning for

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[1]Aitor López-Sánchez is with CETINIA, University Rey Juan Carlos, Spain and Centrale Lille– Univ. Lille, CNRS, Inria, UMR 9189 CRIStAL, France aitor.lopez@urjc.es

[2*] (Corresponding author) Marin Lujak is with CETINIA, University Rey Juan Carlos, Spain marin.lujak@urjc.es

[3]Frederic Semet is with Centrale Lille– Univ. Lille, CNRS, Inria, UMR 9189 CRIStAL, France, frederic.semet@centralelille.fr

[4]Holger Billhardt is with CETINIA, University Rey Juan Carlos, Spain holger.billhardt@urjc.es

multifield areas, incorporating various heterogeneous agricultural machines (tractor-implement combinations) [8], [9]. Lujak et al. [2] introduce the coordination of heterogeneous agribots for complex farming tasks involving coordination in autonomous tractors and implements. Similarly, Ju et al. explore the use of agricultural robots and Multirobot Systems (MRSs) for improved efficiency, joining aerial and ground robots, and manipulators [10].

Distributed solution approaches using multi-agent systems (MAS), where each autonomous vehicle collaborates to achieve a shared mission, are recommended for these problems but remain underexplored [11], [12]. MAS offer scalable, computationally efficient solutions while maintaining the privacy and operational constraints of individual agents (e.g., vehicle goals, limitations, or priorities), making them highly applicable to real-world scenarios.

Agribots and their implements function as a coordinated unit. When an agribot changes its route, the attached implement must follow suit. This demonstrates the reliance of non-autonomous implements on autonomous agribots for navigation. One approach that addresses this interdependence involves allowing implements to detach and reattach during the route, similar to the VRP with Trailers and Transshipments [13], [14] and the Active Passive VRP [15]. This type of synchronization is essential in agricultural operations, where implements are used in conjunction with agribots within specific timeframes. [16], [17]. For example, the Synchronized Sprayer Tanker Routing Problem with Variable Service Time (SSTRPVST) addresses multiple synchronization constraints encountered in precision agriculture, involving both tender tankers and sprayer fleets. To minimize sprayer waiting time and total routing distance, Alkaabneh et al. [18] propose a computationally complex Mixed-Integer Programming (MIP) model. They employ an adaptive large neighborhood search metaheuristic to efficiently solve large-scale instances with up to 60 customers. Another example is in sugarcane field management where Worasan et al. [19] differentiate between two key problems: tractor and tool allocation and scheduling and tractor routing.

The proposed CAFVRPILA model is a continuation of [2], [20], [21], where we proposed models for balancing fairness and efficiency for the (static) multiple TSP and the VRP with fair profits and time windows, introducing the concepts of utilitarian and egalitarian social welfare in vehicle routing. In this paper, we address efficient coordination of heterogeneous agribot and implement routes designed for multiple operations. We emphasize synchronizing the routes of each agribot with its compatible implements.

III. Problem Definition

CAFVRPILA optimizes routes of agribots (vehicles) and their detachable capacitated implements, considering their mutual compatibilities and operational constraints. Our objectives are to minimize the overall fleet costs as well as the costs incurred by the most costly vehicle.

We consider a set of heterogeneous agribots $\mathscr{V}$ and implements $\mathscr{M}$ to perform tasks $\mathscr{K}$, each belonging to an agricultural operation $\mathscr{O}$, (e.g., plowing, seeding, spraying, fumigation, and harvesting).

Each task $k \in \mathscr{K}$ involves performing an agricultural operation o_k on a specific crop field, with an associated demand q_k. This demand represents the quantity of resources that must be either distributed or collected, depending on the nature of the operation. Zero demand represents an operation that does not need any resources (e.g. plowing). The crop fields are designed based on similarities in agricultural parameters, including the crop grown, humidity, sun exposure, soil conditions, and the relative resource efficiency of the size of the field [22]. Different operations (representing different tasks) may be requested for the same crop field.

To perform the tasks $\mathscr{K}^o \subseteq \mathscr{K}$ related with each operation o, there is available a non-empty set $\mathscr{M}_o \subseteq \mathscr{M}$ of implements that can perform them. Each implement m involved in a non-zero demand operation has a resource capacity denoted by Q^m. Implements are initially stored at one of the depots s_m and must be returned to its end depot e_m. For instance, sprayers are tools suitable for multiple tasks such as, e.g., irrigation, herbicide, and pesticide spraying, while plows are employed only for plowing. Even though there are implements that may be used for different operations, the preparation and setup of the implement to switch from one operation to another is usually costly and timely. Therefore, we assume that the same implement may not be used for more than one operation.

The fleet of agribots (vehicles) $\mathscr{V}$ are required to carry the implements and complete the tasks. Each vehicle $v \in \mathscr{V}$ can mount one implement at a time and is compatible with a subset of implements $\mathscr{M}^v \subseteq \mathscr{M}$. Vehicles can switch implements sequentially at the depots. Vehicles begin their routes at their start depot s_v, incur a distance cost C^v per unit of travel, and are constrained by a distance limit D^v based on their autonomy, ensuring they can reach their end depot e_v. The overall fleet's mission is to optimize the fleet's cost while accomplishing all the tasks respecting restrictions regarding implement and vehicle compatibilities as well as task demands and vehicle autonomies.

A. Mathematical formulation

For simplicity and without loss of generality, we assume that the crop fields are located within the farming region, defined by an interconnected transport network. This network is modeled as a complete arc-weighted digraph $\mathscr{G} = (\mathscr{N}, \mathscr{A})$, composed of a set $\mathscr{N}$ of nodes and a set $\mathscr{A}$ of arcs $(i, j) \in \mathscr{N} \times \mathscr{N}$ with $i \neq j$. The set of arcs $\mathscr{A}$ represents the shortest paths between any two distinct nodes $i, j \in \mathscr{N}$ in digraph $\mathscr{G}$. The travel distance for each arc $(i, j) \in \mathscr{A}$ is given by the distance matrix $d_{ij} \geq 0$. Each arc is associated with a travel cost $c^v_{ij} = d_{ij} \cdot C^v$ for vehicle v. Set $\mathscr{N}$ consists of two disjoint sets: the depot set $\mathscr{N}_{depots} = \{s^m, e^m, s^v, e^v \mid m \in \mathscr{M}, v \in \mathscr{V}\}$, containing the start and end depots for implements and vehicles, and the task set $\mathscr{K}$, representing the locations where agricultural operations take place.

To address the problem, we propose a mixed integer linear programming (MILP) model. Based on the parameters out-

TABLE I: CAFVRPILA elements

Symbol	Description
	Fleet's shared information
$\mathscr{K}$	Set of tasks $k, k' \in \mathscr{K}$.
$\mathscr{N}_{depots}$	Set of depots $s, e \in \mathscr{N}_{depots}$.
$\mathscr{N}$	Set of nodes $i, j, h \in \mathscr{N} = \mathscr{N}_{depots} \cup \mathscr{K}$.
$\mathscr{A}$	Set of arcs $(i, j) \in \mathscr{A}$.
$\mathscr{O}$	Set of agricultural operations $o \in \mathscr{O} = \{o_1, \ldots, o_O\}$.
$\mathscr{V}$	Set of vehicles $v \in \mathscr{V} = \{v_1, \ldots, v_V\}$.
$\mathscr{M}$	Set of implements $m \in \mathscr{M} = \{m_1, \ldots, m_M\}$.
$\mathscr{K}^o$	Set of tasks of operation o, $\mathscr{K}^o \subseteq \mathscr{K}$.
$\mathscr{M}_o$	Compatible implements with operation o, $\mathscr{M}_o \subseteq \mathscr{M}$.
	Global parameters
d_{ij}	Distance between i and j.
q_k	Resource demand of task k.
	Implement's information
s^m, e^m	Start and end depot nodes of implement m.
Q^m	Capacity of implement m.
$\mathscr{V}^m$	Set of compatible vehicles with implement m.
	Vehicle's information
s^v, e^v	Start and end depot node of vehicle v.
c^v_{ij}	Cost of traveling arc $(i, j) \in \mathscr{A}$ by vehicle v.
C^v	Vehicle cost per unit distance.
D^v	Autonomy of vehicle v.
$\mathscr{M}^v$	Set of compatible implements with vehicle v.
	Decision variables
x^v_{ij}	Vehicle route variable, $x^v_{ij} = 1$ if vehicle $v \in \mathscr{V}$ travels from vertex i to j, where $(i, j) \in \mathscr{A}$.
ξ^m_{ij}	Implement route variable, $\xi^m_{ij} = 1$ represents if implement $m \in \mathscr{M}$ travels arc $(i, j) \in \mathscr{A}$.
y_i	Distance variable expresses the distance traveled when task i is visited.
z^m_i	Load variable represents the remaining load of an implement m when it leaves node i.

lined in Table I, we define the following decision variables. Binary decision variables x^v_{ij} and ξ^m_{ij} are used for finding the routes of vehicles and implements, respectively. $x^v_{ij} = 1$ ($\xi^m_{ij} = 1$) if vehicle $v \in \mathscr{V}$ (implement $m \in \mathscr{M}$) travels arc $(i, j) \in \mathscr{A}$. Continuous variables y_i denote distance traveled when visiting node $i \in \mathscr{N}$. Finally, continuous variables z^m_i represent the remaining capacity of implement m when leaving node $i \in \mathscr{N}$. The CAFVRPILA mathematical model is presented in (A.1)-(A.17).

$$\min f_{util} = \sum_{v \in \mathscr{V}} \sum_{(i,j) \in \mathscr{A}} c^v_{ij} x^v_{ij} \tag{A.1}$$

$$\min f_{egal} = w \tag{A.2}$$

$$s.t. \sum_{m \in \mathscr{M}_o} \sum_{(i,k) \in \mathscr{A}} \xi^m_{ik} = 1 \qquad \forall o \in \mathscr{O}, k \in \mathscr{K}^o, \tag{A.3}$$

$$\sum_{(i,k) \in \mathscr{A}} \xi^m_{ik} = \sum_{(k,j) \in \mathscr{A}} \xi^m_{kj} \qquad \forall o \in \mathscr{O}, m \in \mathscr{M}_o, j \in \mathscr{N} \setminus \{e^m\}, \tag{A.4}$$

$$\sum_{(s^m,j) \in \mathscr{A}} \xi^m_{(s^m)j} \leq 1 \qquad \forall o \in \mathscr{O}, m \in \mathscr{M}_o, \tag{A.5}$$

$$z^m_j - z^m_i \geq q_j - Q^m(1 - \xi^m_{ij}) \qquad \forall o \in \mathscr{O}, m \in \mathscr{M}_o, (i,j) \in \mathscr{A}, \tag{A.6}$$

$$0 \leq z^m_i \leq Q^m \qquad \forall o \in \mathscr{O}, m \in \mathscr{M}_o, i \in \mathscr{N}, \tag{A.7}$$

$$\xi^m_{ij} \in \{0,1\} \qquad \forall m \in \mathscr{M}, (i,j) \in \mathscr{A}, \tag{A.8}$$

$$\xi^m_{ij} \leq \sum_{v \in \mathscr{V}^m} x^v_{ij} \qquad \forall o \in \mathscr{O}, m \in \mathscr{M}_o, (i,j) \in \mathscr{A}, \tag{A.9}$$

$$\sum_{(i,j) \in \mathscr{A}} x^v_{ij} = \sum_{(j,h) \in \mathscr{A}} x^v_{jh} \qquad \forall v \in \mathscr{V}, j \in \mathscr{N} \setminus \{e^v\}, \tag{A.10}$$

$$\sum_{(s^v,j) \in \mathscr{A}} x^v_{s^v j} \leq 1 \qquad \forall v \in \mathscr{V}, \tag{A.11}$$

$$\sum_{(i,j) \in \mathscr{A}} d_{ij} x^v_{ij} \leq D^v \qquad \forall v \in \mathscr{V}, \tag{A.12}$$

$$y_j \geq y_i + d_{ij} - D^v(1 - x^v_{ij}) \qquad \forall v \in \mathscr{V}, (i,j) \in \mathscr{A}, \tag{A.13}$$

$$y_i \geq 0 \qquad \forall i \in \mathscr{N}, \tag{A.14}$$

$$x^v_{ij} \in \{0,1\} \qquad \forall v \in \mathscr{V}, (i,j) \in \mathscr{A}, \tag{A.15}$$

$$\sum_{(i,j) \in \mathscr{A}} c^v_{ij} x^v_{ij} \leq w \qquad \forall v \in \mathscr{V}, \tag{A.16}$$

$$w \in \mathbb{R}. \tag{A.17}$$

The modeling of the objective function depends on the problem's specific goals. If the aim is to minimize the overall cost of vehicles and implements, aligning with a utilitarian social welfare approach, the model is defined by equations (A.1), (A.3)-(A.15). Alternatively, if the objective is to ensure fair assignment by minimizing the cost incurred by the highest cost vehicle (worst-off vehicle) according to egalitarian social welfare, it is expressed by the equations (A.2)-(A.17).

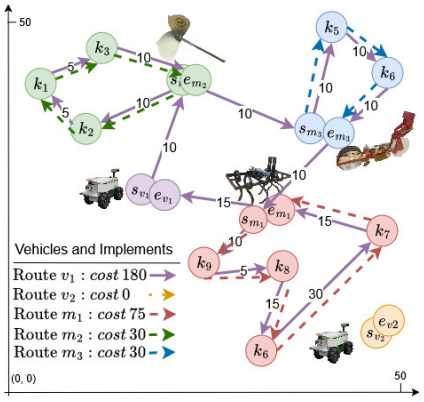
Constraints (A.3)-(A.8) are the implement routing constraints. Specifically, constraints (A.3) guarantee that each task must be performed by an implement compatible with its operation; constraints (A.4) state that if an implement visits a node, it should also depart from it; constraints (A.5) ensure that each implement is used at most once; constraints (A.6) and (A.7) ensure that the demands of the tasks visited by any implement $m \in \mathscr{M}$ do not exceed its capacity Q^m; and finally, (A.8) are the implement routing decision variables.

Constraints (A.10)-(A.15) are the vehicle routing constraints. Constraints (A.10) state that if a vehicle visits a node, it should also leave it; constraints (A.11) ensure that each vehicle is used at most once; constraints (A.12) ensure that the travel distance for any vehicle $v \in \mathscr{V}$ does not exceed its autonomy D^v; constraints (A.13) enforce that the distance traveled from a node is greater than the distance traveled from the previous node plus the distance between the two nodes; lastly, constraints (A.15) define the vehicle routing decision variables. Finally, the coordination constraints (A.9) between vehicles and implements assure that if an implement travels along an arc, it must be attached to a compatible vehicle.

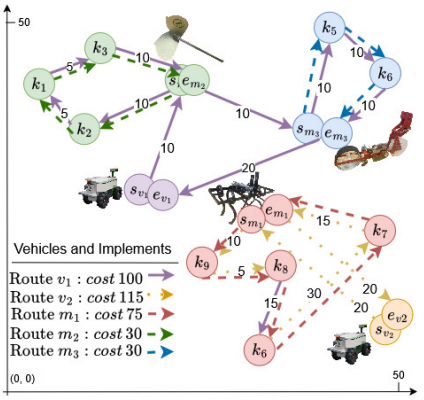
Use case example. Figure 1 highlights how different optimization objectives influence route planning in a scenario where a fleet of two vehicles must complete nine tasks having available three compatible implements. Figure 1a shows the solution that maximizes the utilitarian social welfare (minimizes the overall sum of the fleet's costs) and Figure 1b maximizes the egalitarian social welfare (minimizes the individually highest agribot cost).

IV. Multi-agent Solution Approach

Traditional heuristics mostly consider a fixed robot-implement pair and overlook that the same agribot can switch implements during operation. To improve efficiency, inspired by the Active-Passive VRP approach in [15], we consider vehicle-implement reconfigurability and propose a two-phase route optimization hierarchical model. We first



(a) Utilitarian social welfare solution.



(b) Egalitarian social welfare solution.

Fig. 1: Use case example.

find implements' routes and then the routes of agribots following the precalculated implements' routes. We do so since, usually, implement capacity is more restrictive than the agribot autonomy. While electric agribots can operate for 8–12 hours on a single charge, implements like seeders or sprayers usually require refilling every 2–4 hours.

The CAFVRPILA problem combines capacitated implement routing, limited agribot autonomy, and agribot-implement synchronization. Given that this problem is intrinsically spatially distributed and computationally expensive, we introduce a multi-agent heuristic approach, decomposing the problem into the subproblems resolved separately by implement and agribot agents. Our decomposed model follows the following formulation: constraints (A.3)-(A.8) govern implement routes per operation, constraints (A.10)-(A.14) define agribot routes, and coordinating constraints (A.9) ensure each implement is attached to an agribot.

A. Phase 1: Operation decomposition and implement routes

In Phase 1, the objective is to obtain implement routes that complete agricultural tasks. Each operation can be modeled as an agent that is willing to find a set of at most $|\mathscr{M}_o|$ implement routes, ensuring each task in the set $\mathscr{N}^o_{tasks}$ is performed once without exceeding implement capacity.

For each operation agent, let $\mathscr{G}^o = (\mathscr{N}^o, \mathscr{A}^o)$ denote the subgraph of $\mathscr{G}$ induced by the tasks $\mathscr{N}^o_{tasks}$ and depots $\mathscr{N}^o_{depots} = \{s^m, e^m \,|\, m \in \mathscr{M}_o\}$ compatible with operation o. Next, we propose two different objective functions for generating implement routes: 1. minimizing the overall distance traveled by implements as expressed in $P1^o_{util}$: (B.1), (B.3)-(B.9); 2. minimizing the maximum distance of implements, denoted by L, to achieve fairness between them in $P1^o_{egal}$: (B.2), (B.3)-(B.9).

$$P1^o_{util}: \min \sum_{m\in\mathscr{M}_o}\sum_{(i,j)\in\mathscr{A}^o} d_{ij}\xi^m_{ij} \tag{B.1}$$

$$P1^o_{egal}: \min L \tag{B.2}$$

$$s.t. \sum_{m\in\mathscr{M}_o}\sum_{(i,k)\in\mathscr{A}^o} \xi^m_{ik} = 1 \qquad \forall k \in \mathscr{N}^o_{tasks}, \tag{B.3}$$

$$\sum_{(i,k)\in\mathscr{A}^o} \xi^m_{ik} = \sum_{(k,j)\in\mathscr{A}^o} \xi^m_{kj} \qquad \forall m \in \mathscr{M}_o,\ j \in \mathscr{N}^o \backslash \{e^m\}, \tag{B.4}$$

$$\sum_{(s^m,j)\in\mathscr{A}^o} \xi^m_{(s^m)j} \le 1 \qquad \forall m \in \mathscr{M}_o, \tag{B.5}$$

$$z^m_j - z^m_i \ge q_j - Q^m(1-\xi^m_{ij}) \qquad \forall m \in \mathscr{M}_o,\ (i,j) \in \mathscr{A}^o, \tag{B.6}$$

$$0 \le z^m_i \le Q^m \qquad \forall m \in \mathscr{M}_o,\ i \in \mathscr{N}^o, \tag{B.7}$$

$$\xi^m_{ij} \in \{0,1\} \qquad \forall m \in \mathscr{M}_o,\ (i,j) \in \mathscr{A}^o, \tag{B.8}$$

$$\sum_{(i,j)\in\mathscr{A}^o} d_{ij}\xi^m_{ij} \le L \qquad \forall m \in \mathscr{M}_o. \tag{B.9}$$

By solving the previous models for each operation o, the agent obtains a set of implement routes that visit the tasks in the set $\mathscr{N}^o_{tasks}$ associated with the operation. Let the solution set $\mathscr{S}^o_{util}$ represent the decision variable values $\{\xi^m_{ij} \,|\, m \in \mathscr{M}_o, (i,j) \in \mathscr{A}^o\}$ obtained from $P1^o_{util}$, and let $\mathscr{S}^o_{egal}$ denote those obtained from $P1^o_{egal}$. By aggregating the solutions for each operation, the utilitarian route solution that visits all tasks is defined as $\mathscr{S}_{util} = \{\xi^m_{ij} \,|\, o \in \mathscr{O}, m \in \mathscr{M}_o, (i,j) \in \mathscr{A}^o\}$. Similarly, the egalitarian route solution is denoted by $\mathscr{S}_{egal}$.

Once the implement route sets $\mathscr{S}_{util}$ and $\mathscr{S}_{egal}$ are obtained, it remains for these routes to be assigned to the vehicles so that they can be carried out. The overall cost of the system will depend on the position of implements and vehicles and also the autonomy of the vehicle fleet, potentially assigning multiple implement routes to the same vehicle while minimizing the overall cost.

B. Phase 2: Vehicle routes

During this phase, vehicle agents must divide up the Phase 1 implement routes based on their compatibilities. Each vehicle, initially stationed at its depot, must travel to the implement depots, attach the implements, and execute the designated routes. The assignment of each agent can be done considering the utilitarian social welfare model:

$$P2_{util}: \min \sum_{v\in\mathscr{V}}\sum_{(i,j)\in\mathscr{A}} c^v_{ij}x^v_{ij} \tag{C.1}$$

$$s.t. \quad \text{(A.9), (A.10)} - \text{(A.15)}. \tag{C.2}$$

Or by the egalitarian social welfare model:

$$P2_{egal}: \min \quad w \tag{C.3}$$

$$s.t. \quad \text{(A.9), (A.10)} - \text{(A.17)}. \tag{C.4}$$

TABLE II: Definition of the instances

Instance Name	Implements Capacity	#Depots	#Vehicles	Vehicles Autonomy	#Depots
1	100	1DI	10	1000	1DV
2	100	1DI	10	1000	+DV
3	100	+DI	10	1000	1DV
4	100	+DI	10	1000	+DV
5	200	1DI	10	1000	1DV
6	200	1DI	10	1000	+DV
7	200	+DI	10	1000	1DV
8	200	+DI	10	1000	+DV
9	500	1DI	10	1000	1DV
10	500	1DI	10	1000	+DV
11	500	+DI	10	1000	1DV
12	500	+DI	10	1000	+DV
13	1000	1DI	10	1000	1DV
14	1000	1DI	10	1000	+DV
15	1000	+DI	10	1000	1DV
16	1000	+DI	10	1000	+DV

Because the values of ξ_{ij}^{m} were calculated in Phase 1, they are considered as the parameters of the formulation. This leaves 4 possible combinations solving $P2_{util}(\mathscr{S}_{util})$, $P2_{util}(\mathscr{S}_{egal})$, and solving $P2_{egal}(\mathscr{S}_{util})$ with $P2_{egal}(\mathscr{S}_{egal})$.

V. Computational results

In this section, we evaluate the two-phase decomposition heuristic through functional experiments on 16 distinct instances, representing a range of operational scenarios. The proposed models are implemented in Python with Gurobipy and OR-Tools [23]. Simulations are conducted on a virtualized cluster featuring an Intel Xeon Gold 6226R processor[1], equipped with 16 CPUs, 32 GB of RAM, and 2.9 GHz.

a) Experiment setup: The proposed CAFVRPILA model lacks an established benchmark. To evaluate its performance, our study consists of 16 instances with 5 operations each one, which are derived from the Augerat instances "A-n80-k10.vrp", "A-n63-k10.vrp", "A-n62-k8.vrp", "A-n60-k9.vrp", and "A-n53-k7.vrp" [24]. Each Augerat instance represents a different set of tasks with their compatible implements (a maximum of 10 with varying capacities). The instances[2] contain a fleet of 10 vehicles, each with a maximum autonomy of 1000 units, and include 383 nodes, 323 tasks, 50 implement depots, and 10 depots.

To create diverse experimental scenarios, each of the 16 instances is defined by different parameter configurations. Two vehicle depot scenarios are considered: one where all vehicles share a single depot location ($1DV$) and another where each vehicle has its own randomly distributed depot ($+DV$). Similarly, we explore two implement depot configurations: one where all implements for a given operation share a common depot ($1DI$), and another where each implement has a dedicated depot ($+DI$). Additionally, implement capacities are varied across four levels—100, 200, 500, and 1000—to simulate different operational conditions in the field.

[1]https://www.intel.es/content/www/es/es/products/sku/199347/intel-xeon-gold-6226r-processor-22m-cache-2-90-ghz/specifications.html

[2]https://github.com/aitorls/CAFVRPILA-instances.git

TABLE III: Phase 1 results obtained by utilitarian and egalitarian social welfare models of implement routes.

Instance	Implement Routing				
$P1^{o}_{util}:\mathscr{S}_{util}$	Dist	#I	Max	Min	Mean
1,2	6991	43	294	42	162.5
3,4	3801	45	137	6	84.4
5,6	4524	24	305	10	188.5
7,8	3167	36	181	4	87.9
9,10	3319	10	452	172	331.9
11,12	3018	26	343	4	116.0
13,14	3035	5	724	553	606.6
15,16	3275	17	707	4	174.8
Mean	3854.3	25.8	394.0	99.4	219.2
$P1^{o}_{egal}:\mathscr{S}_{egal}$	Dist	#I	Max	Min	Mean
1,2	7829	45	249	50	174.0
3,4	4318	50	119	6	86.4
5,6	5876	29	249	16	202.6
7,8	3840	45	106	6	85.3
9,10	5496	27	249	50	203.6
11,12	3907	47	105	6	83.1
13,14	5666	28	249	50	202.4
15,16	3927	46	106	6	85.4
Mean	5114.7	39.7	179.1	23.8	140.3

b) Analysis of the implement routes: The results of implement routes obtained from solving the two models $P1^{o}_{util}$ and $P1^{o}_{egal}$ are summarised in Table III. This table provides information on the overall distance (Dist) traveled by implements, the number of implements used (Implements), and the maximum, minimum, and average distance for each implement (Max, Min, Mean), of the set of generated implement routes,$\mathscr{S}_{util}$ and $\mathscr{S}_{egal}$.

In the utilitarian implement solution $\mathscr{S}_{util}$, instances with lower vehicle capacity (1 to 8) use more implements due to their low capacity, resulting in more frequent trips to depots, shorter average distances per implement, and a higher total distance. Interestingly, when each implement has its own independent depot ($+DI$), the total distance is reduced because the depots are spread out across the region. Also, instances with high vehicle capacity $(500, 1000)$ require fewer implements overall, and with a single depot ($1DI$), this number decreases to just one implement per operation, as seen in instances 13 and 14.

Comparing the egalitarian solution ($P1^{o}_{\text{egal}}$) to the utilitarian one ($P1^{o}_{\text{util}}$), the egalitarian approach achieves a fairer distribution of values by reducing the amplitude of the variations. This is accomplished by using a greater number of implements, which in turn lowers the average cost per vehicle and decreases the disparity in distances between implements. While the worst-off implement distance is cut by 45% compared to the utilitarian solution, the total distance increases by 1.3 times. These differences are particularly evident in high-capacity instances (9 to 16), where the greatest reduction in the worst-off implement distance occurs. In the utilitarian model, higher capacity enables longer routes, reducing total distance but compromising fairness.

c) Analysis of the vehicle routes: Table IV presents the results of the four combinations of vehicle route allocation models in Phase 2: $P2_{util}(\mathscr{S}_{util})$, $P2_{util}(\mathscr{S}_{egal})$, $P2_{egal}(\mathscr{S}_{util})$

TABLE IV: Phase 2 results obtained by the all combination model of vehicle routes.

Instance name	$P2_{util}(\mathscr{S}_{util})$			$P2_{util}(\mathscr{S}_{egal})$			$P2_{egal}(\mathscr{S}_{util})$			$P2_{egal}(\mathscr{S}_{egal})$		
	Cost	#V	Max	Cost	#V	Max	Cost	#V	Max	Cost	#V	Max
1	**8133**	9	992	9103	10	990	8357	10	**845**	9160	10	922
2	**7685**	9	1000	9027	10	996	8122	10	**818**	9141	10	917
3	**4354**	5	985	4898	6	982	4808	10	**485**	5288	10	544
4	**4160**	7	943	4670	7	963	4515	10	**457**	5060	10	514
5	**5283**	6	989	6936	8	994	5871	10	**617**	7456	10	798
6	**5043**	6	954	6635	7	999	5664	10	**581**	7069	10	756
7	**3703**	4	999	4394	5	988	4197	10	**431**	4788	10	492
8	**3521**	6	983	4199	7	976	3868	10	**408**	4496	10	480
9	**3927**	5	938	6876	8	996	4461	9	**602**	6955	10	769
10	**3611**	5	916	6253	7	1000	4239	9	**538**	6582	10	709
11	**3480**	4	981	4471	5	1000	3964	10	**405**	4960	10	506
12	**3364**	7	964	4244	7	946	3701	10	**384**	4503	10	475
13	**3641**	5	874	6726	8	903	**3641**	5	874	7117	10	**778**
14	**3325**	5	852	6425	7	1000	3415	5	810	6855	10	**740**
15	**3435**	5	899	4403	5	972	3494	5	874	4904	10	**506**
16	**3275**	6	978	4286	7	978	3292	6	739	4560	10	**481**
Mean	4395.9	5.9	952.9	5846.6	7.1	980.2	4720.1	8.7	609.6	6180.9	10.0	649.2

and $P2_{egal}(\mathscr{S}_{egal})$, where implement routes $\mathscr{S}_{util}$ and $\mathscr{S}_{egal}$ are generated in Phase 1. The table reports overall cost (Cost), number of vehicles used (#V), and the cost of the worst-off vehicle (Max).

Regardless of the model used, the number of vehicle depots impacts overall cost. The utilitarian model ($P2_{util}$) reduces vehicle use across both implement route solutions ($\mathscr{S}_{util}$ and $\mathscr{S}_{egal}$), minimizing vehicle costs by reducing trips to and from depots. Using $P2_{util}(\mathscr{S}_{util})$ results in better overall costs and fewer vehicles, with similar worst-off vehicle costs. However, $P2_{util}(\mathscr{S}_{egal})$ does not improve either overall cost or the worst-off vehicle's cost. The egalitarian models aim to distribute routes more fairly, particularly in $P2_{egal}(\mathscr{S}_{egal})$, where all 10 vehicles are used. The model $P2_{egal}(\mathscr{S}_{util})$ achieves better overall cost and fairness in instances 1-12, where implement routing is capacity-constrained. In instances with greater vehicle autonomy (13-16), $P2_{egal}(\mathscr{S}_{egal})$ significantly reduces the worst-off vehicle's cost, although at a higher total cost.

Therefore, the utilitarian model $\mathscr{S}_{util}$ tends to produce large routes in unconstrained capacity cases, whereas shorter implement routes in $\mathscr{S}_{egal}$ allow for fairer vehicle assignments. Optimizing shorter implement routes could further balance fairness and efficiency in $P2_{egal}$, a potential improvement explored in the following experiments.

d) Improving the egalitarian social welfare solution: To achieve fairness and minimize the cost of the worst-off vehicle in $P2_{egal}$, a trade-off must be achieved between the individual deployment route distances and the total distance. Instead of minimizing L in the $P1^{o}_{egal}$ model, fairness is promoted by adjusting the parameter L of constraint (B.9) in the $P1^{o}_{util}$ model. This adjustment results in shorter implement routes while maintaining minimized overall distance. To validate this approach, we varied the parameter L from 100 to 1000 in increments of 100 for each operation in the P^{o}_{util} model. We then solved $P2_{egal}$ and compared the worst-off vehicle's minimum cost to the results obtained without limiting L. The results, summarized in Table V, show that this method improves both the "Max" column (worst-off cost) and reduces overall cost, especially when compared to $P2_{egal}(\mathscr{S}_{egal})$ in instances 13 to 16.

TABLE V: Results for the egalitarian social welfare model varying the maximum distance of implement routes.

Instance name	Varying L Vehicles Cost		Max	Instance name	Varying L Vehicles Cost		Max
1	**8357**	10	**845**	9	4536	10	**547**
2	**8122**	10	**818**	10	4370	10	**526**
3	**4808**	10	**485**	11	**3964**	10	**405**
4	**4515**	10	**457**	12	**3617**	10	**384**
5	5967	10	**611**	13	4511	10	**546**
6	**5649**	10	**581**	14	4265	9	**526**
7	**4197**	10	**431**	15	4081	10	**425**
8	3955	10	**405**	16	3686	10	**394**

VI. Conclusions

This study proposed the distributed multi-agent based CAFVRPILA model for routing cooperative fleets composed of agribots with implements. Our proposed 2-phase approach included implement routing in the first phase, focusing on maximizing utilitarian social welfare. In the second phase, we considered both egalitarian and utilitarian social welfare when optimizing vehicle routing. The analysis of the implement routes revealed that instances with limited capacity tend to use more implements and the spatial distribution of implement depots significantly affects overall cost. Strategically dispersing implement depots across the region improves task distribution and reduces total distance.

Utilitarian models, which focused on minimizing overall distance, employed fewer implements, reducing the total distance. Conversely, egalitarian models distributed the distance across implements, increasing fairness but at the expense of higher total distance, particularly in high-capacity instances.

Utilitarian vehicle and implement model consistently reduced vehicle usage, resulting in cost savings. Instances with independent vehicle depots used more vehicles, yet achieved better cost efficiency and fairness compared to those with shared depots. However, the egalitarian vehicle model failed to balance results across all implement routing solutions. While the utilitarian implement model yielded lower costs for the worst-off vehicle in low-capacity instances, it did not guarantee fair assignment in high-capacity scenarios.

To improve both fairness and efficiency, a compromise between individual implement route distance and overall distance is needed by adjusting the parameter L. Experimental validation through parameter variation showed improved overall cost and a reduction in the worst-off vehicle's cost (especially in high-autonomy instances), effectively reducing costs and enhancing fairness, demonstrating its potential.

This work was performed in experiments focusing on a multi-agent based architecture for a Unity3D simulation of dynamic agribots fleet coordination [25]. In future work, we plan to implement the CAFVRPILA model within a real agribot fleet and consider a dynamic, open fleet where problem data may evolve throughout the mission. This includes accommodating the dynamic appearance and removal of both agribots and tasks during the planning horizon. To enhance planning, we will explore receding horizon control across multiple decision periods, which will allow us to adjust for fairness and support disadvantaged agribots in later periods.

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